

- discrete time: random walk, markov process
- continuous time: brownian motion, stopping times, martingales
- Ito / stochastic calculus
- Risk neutral probability measure
- Risk neutral valuation, black scholes
- Tricks to avoid black scholes, law of unconsciousness statistician (=fundamental theorem of asset pricing)
- Call prices 'span' all derivative prices
- solving black scholes with neural networks
- the problem of trading frequency vs model accuracy

1 Risk neutral evaluation of derivatives

. A derivative (portfolio) is a collection of assets. The question is how to price such a derivative. Assets

References